# Partner-Specific Affective Precision in Social Active Inference

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**Abstract.** Affective active inference formalizes valence as inference over expected policy precision: an internal estimate of how well an agent's generative model supports action. In social interaction, however, model fitness is not only global. An agent may predict one partner reliably, remain uncertain about another, and be misled by a third. We extend this formulation by factorizing affective precision across partners in a repeated trust-game simulation. The agent maintains separate predictive models for each partner; after each interaction, partner-specific prediction error updates a local precision estimate, which modulates confidence in future policies involving that partner.

We evaluate stable interaction, partner choice, and abrupt betrayal regimes. Partner-local precision changes how social beliefs are acted upon, shaping which partners the agent returns to, how decisively it acts, and how it reallocates after surprising outcomes. These effects are strongest in decision-ambiguous regimes, where multiple actions remain viable. They can also emerge even when agents with and without precision modulation maintain similar beliefs about their partners, indicating a deployment rather than knowledge difference. Because this affective signal tracks predictive reliability more than partner reward, its behavioral influence supports a model-fitness interpretation. Under abrupt betrayal, however, surprise-driven precision updates can over-sharpen decisions, locking the agent into outdated assumptions and reducing payoffs after the behavioral switch. These results suggest that extending affective inference to multi-agent systems requires localizing precision over social models, providing a computational account of how implicit metacognitive estimates of model fitness can both support and misdirect social action.

**Keywords:** Active inference · Affect · Trust games

## 1 Introduction

Active inference treats perception and action as coupled inference under a generative model [3, 4, 9]. Agents evaluate policies in terms of expected free energy, balancing pragmatic outcomes with epistemic value [5]. Policy precision then controls how decisively an agent acts on the currently preferred policy rather than preserving a diffuse distribution over plausible actions.

Affective active-inference accounts give this precision psychological content. Hesp et al. [7] formalize valence as an inference over expected action precision or subjective model fitness: positive affect signals that the current generative model supports confident action, while negative affect signals model failure or uncertainty. This makes affect a metacognitive estimate of whether the agent can rely on its own model, rather than an externally added reward bonus.

Social interaction makes this estimate naturally local. A focal agent may predict one partner reliably, remain uncertain about another, and be confidently adversarial toward a third. A single global precision signal collapses these distinctions. In a social setting, the relevant quantity is often not whether the environment is predictable overall, but whether the agent's model of a particular partner is reliable enough to guide action. A predictable exploiter can be low value but high model fitness; a volatile ally can be high value but low model fitness.

We test this idea in a repeated multi-partner trust-game simulation. The agent maintains separate predictive models for each partner. After each interaction, partner-specific prediction error updates a local beta state, which modulates policy precision for future choices involving that partner. The resulting account makes three claims. First, partner-local precision should be behaviorally visible mainly when the policy space is open enough for precision to matter. Second, the signal should track predictive reliability more than realized reward. Third, because it gates deployment, it can help or harm depending on whether the partner model is calibrated to the current social regime.

The contribution is a partner-specific extension of affective precision for social active inference. It is not a claim that affect always improves payoff, nor a claim that beta is a literal trust scalar. The experiments show that partner-local precision changes how social beliefs are acted upon, including partner selection and post-betrayal reallocation, while also revealing a boundary condition: abrupt betrayal can make surprise-driven precision updates over-sharpen decisions around outdated assumptions.

## 2 Model and Experiments

We evaluate partner-specific affective precision in a repeated trust game derived from the economic trust-game tradition [1, 8]. A focal active-inference agent interacts with four partners. Each partner has a latent type (cooperator, reciprocator, exploiter, or random) and a latent stance toward the agent (trusting, neutral, or hostile). The agent observes the partner's action and its own payoff, and it selects either a binary action or a graded investment depending on the regime.

For each partner $k$, the focal agent maintains a separate discrete active-inference model over partner type, partner stance, and own-action bookkeeping. State and policy inference use the standard discrete active-inference machinery implemented in `pymdp` [4, 2, 6]. Partner choice is handled by evaluating partner-local policies and selecting among the resulting candidate interactions.

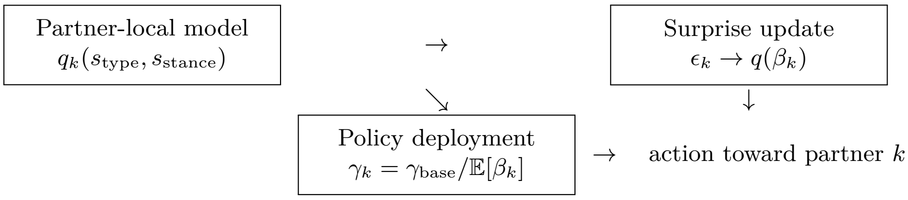



Fig. 1: Partner-specific affective precision. Each partner has a local generative model; prediction error updates a partner-local beta estimate, which modulates the precision of future policies involving that partner.

The affective component is external to the POMDP state space. After interacting with partner $k$, the agent computes partner-action surprise

$$\epsilon_k = 1 - P(o_k^{\text{action}} = o_k^{\text{obs}} \mid q(s_{\text{type}}, s_{\text{stance}})). \tag{1}$$

The tracker converts this into affective charge

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2), \tag{2}$$

following the interpretation of affect as expected action precision and model fitness [7]. A categorical beta posterior $q(\beta_k)$ is updated over discrete beta levels. The expected beta then modulates policy precision for future choices involving that partner:

$$\gamma_k = \gamma_{\text{base}}/\mathbb{E}_{q(\beta_k)}[\beta_k]. \tag{3}$$

Lower beta therefore sharpens the policy posterior; higher beta softens it. A tracked-only lesion keeps the beta update but fixes $\gamma_k = \gamma_{\text{base}}$, isolating deployment from belief updating.

The experiments test five readouts. H0 asks whether policy openness gates visible affect effects. H1 tests whether precision tracks predictive reliability rather than realized reward. H2 compares affective agents with tracked-only lesions to test deployment with similar belief readouts. H3 introduces an abrupt hostile stance switch to test stress and misdeployment. H4 tests whether partner-local precision changes partner selection. H5 perturbation variants are treated as supplemental checks on beta dynamics rather than clinical validation.

## 3 Results

### 3.1 Partner-local precision tracks model fitness

The H1 reliability-versus-reward confirmation is the central mechanism result. Across 30 seeds per variant and 200 rounds per seed, affective precision did not improve payoff: total payoff was 534.6 for the precision-modulated agent and 542.1 for the no-affect baseline, with a mean difference of $-7.53$ and bootstrap CI $[-27.88, 13.82]$. This rules out a simple reward-booster interpretation.

Table 1: Evidence hierarchy used in the paper. H1 and H3 are the primary statistical anchors because they use 30 seeds per variant.

| Card | Mechanism role | Seeds | Status |
|---|---|---|---|
| H0 | Policy openness gates visible precision effects | 5/variant | Supporting |
| H1 | Precision tracks model fitness rather than reward | 30/variant | Primary |
| H2 | Precision changes deployment despite similar beliefs | 5/variant | Supporting |
| H3 | Abrupt betrayal exposes misdeployment risk | 30/variant | Primary |
| H4 | Precision changes partner choice | 5/variant | Supporting |
| H5 | Perturbations separate beta dynamics | 5/variant | Supplemental |

Despite the absence of a payoff advantage, the beta-derived precision signal tracked predictive reliability more strongly than reward. Across partner-seed units, $|\mathrm{corr}(\mathrm{precision}, \mathrm{surprise})| = 0.701$, while $|\mathrm{corr}(\mathrm{precision}, \mathrm{payoff})| = 0.419$. The corresponding seed-level reliability-over-reward summary was $+0.096$ with bootstrap CI $[0.027, 0.164]$. This dissociation supports the model-fitness interpretation: the affective signal estimates confidence in the partner model, not partner value.

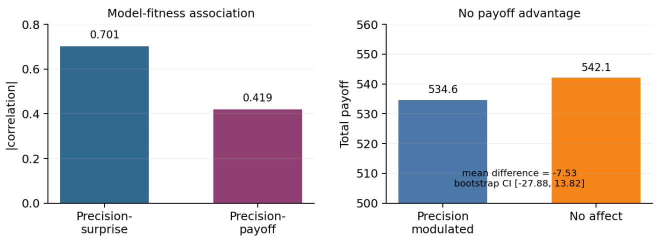



Fig. 2: Model-fitness readout. In the 30-seed H1 confirmation, partner-local precision tracks predictive surprise more strongly than realized payoff, while precision modulation has no total-payoff advantage.

### 3.2 Precision gates deployment in open policy regimes

In the open graded regime, using five seeds per variant, partner-local precision changed behavior in the predicted direction. The precision-modulated agent reached total payoff 1884.6 with mean policy entropy 7.94, while the no-affect or tracked-only baseline reached 1859.4 with entropy 8.80. The difference was therefore $+25.2$ payoff and $-0.86$ entropy for precision modulation.

The tracked-only lesion clarifies the deployment claim. In the same open regime, the precision-modulated agent had joint belief accuracy 0.336 and stance accuracy 0.816, while the tracked-only lesion had joint accuracy 0.339 and stance accuracy 0.855. Belief readouts were therefore similar or slightly favored the lesion, but policy entropy and payoff favored the precision pathway. This is the intended deployment dissociation: the signal changes how beliefs are acted upon rather than simply improving belief accuracy.

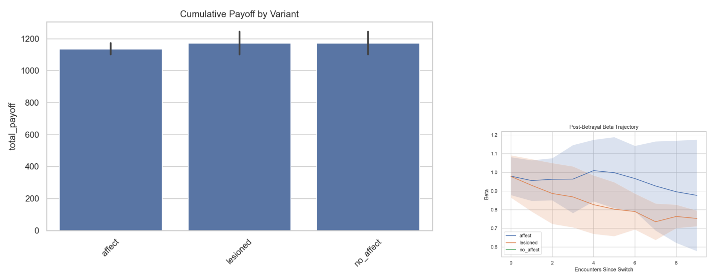
### 3.3 Partner choice changes before payoff

In the H4 partner-choice run, payoff was nearly flat across five seeds: 393.6 for precision modulation versus 393.2 for no affect. The behavioral distribution nevertheless changed. Mean policy entropy was 3.99 with precision modulation and 4.83 without it. Selection shifted toward the first partner (0.348 to 0.366) and away from the third partner (0.162 to 0.138). This supports the social-choice prediction in a bounded way: partner-local precision can shape approach and avoidance before total payoff becomes an informative readout.

### 3.4 Abrupt betrayal exposes a boundary condition

The H3 confirmation shows that the same precision channel can be harmful under abrupt social volatility. Across 30 seeds per variant and 120 rounds per seed, the precision-modulated agent obtained lower whole-run payoff than no-affect or tracked-only baselines: 1136.1 versus 1172.1, with mean difference $-36.08$, bootstrap CI $[-63.65, -10.93]$, and pairwise $p = 0.0169$. Policy entropy was lower under precision modulation (8.38 versus 8.74), showing that the channel was behaviorally active.

Post-switch readouts explain the failure mode. Precision modulation reduced mean reencounters with the switched partner (4.4 versus 6.1) and lowered reencounter selection rate (0.049 versus 0.067), but it did not improve the safety of returning: payoff on reencounter was 8.76 versus 8.91, and wrong-type rate on reencounter was 0.237 versus 0.165. Abrupt betrayal therefore does not show affective recovery. It shows precision-driven misdeployment: decisions become sharper before the partner model has recalibrated enough.




(a) Cumulative payoff (b) Beta trajectory

Fig. 3: Betrayal boundary condition. Abrupt betrayal makes partner-local precision behaviorally visible, but not safer: precision modulation lowers entropy and changes reallocation while reducing whole-run payoff.

### 3.5 Shock shape matters more than generic caution

The precision-sensitivity follow-up asks whether betrayal failure is simply excessive precision gain. In the abrupt sensitivity run, the no-affect/tracked-only baseline had payoff 1153.6 and entropy 8.80. Default precision modulation reached payoff 1140.4 and entropy 8.35 ($p = 0.370$ against no affect). A combined-caution variant raised entropy to 9.27 but lowered payoff to 1115.0 ($p = 0.003$ against no affect).

The gradual betrayal run changed the outcome. The no-affect/tracked-only baseline had payoff 1148.9 and entropy 8.76, while default precision modulation nearly matched it with payoff 1147.6, entropy 8.38, and $p = 0.906$. Combined caution again reduced payoff (1118.3, $p = 0.005$). Generic caution therefore does not rescue abrupt betrayal. The more informative boundary is shock shape: default partner-local precision is most harmful when a partner changes abruptly.

## 4 Discussion

These simulations support a constrained claim: affective precision can be factorized over social partners and used as a local model-fitness signal for social policy deployment. The signal is not equivalent to reward. In the H1 confirmation, it tracks predictive surprise more strongly than payoff while producing no payoff advantage. It is also not merely belief accuracy. In the H2 lesion, belief readouts remain similar while policy entropy and payoff differ. Finally, it is not uniformly beneficial. The H3 betrayal results show that abrupt volatility can make the same precision channel over-sharpen outdated assumptions.

This places the model between two simpler accounts. A reward-cache account cannot easily explain why precision follows prediction reliability more than reward, or why a predictable exploiter may still support confident action. A belief-only account cannot explain why similar partner-belief readouts can produce different policy entropy and payoff. Partner-local affective precision instead acts as a metacognitive deployment variable: it summarizes whether the agent's model of a particular partner is currently reliable enough to act on.

The work complements recent multi-agent active-inference models that factorize beliefs or reason about other agents [11, 10]. Our focus is different. We do not introduce a general theory-of-mind planning algorithm, nor do we assume that all agents share a generative model. We isolate a local precision channel that regulates how a focal agent deploys its own partner-specific social models. This is also distinct from active-inference accounts of trust in human-machine interaction [12]: here, the precision signal is not a literal trust scalar but confidence in the agent's model of a partner.

*Limitations.* The evidence is simulation-only and should not be read as validation of human affect or clinical phenotypes. The primary partners are scripted, which isolates the focal agent's mechanism but does not establish fully reciprocal multi-agent dynamics. H0, H2, and H4 currently use five seeds per variant and should be read as supporting evidence; H1 and H3 are the stronger 30-seed anchors. Finally, the architecture has not yet been compared against a global-beta ablation. That comparison is the most direct next test of whether partner-specific affect is necessary, rather than simply useful.

*Future work.* The most important follow-up is a global-beta control that shares one affective precision estimate across partners while preserving the same beta-to-policy-precision mapping. A second follow-up is a focused shock-shape gradient that varies betrayal abruptness while holding the default precision channel fixed. More broadly, partner-local affective precision should be tested in richer reciprocal environments and, eventually, against human behavior.

## 5 Conclusion

Partner-specific affective precision provides a compact mechanism for linking social model fitness to action. In repeated trust-game simulations, it tracks predictive reliability more than partner reward, changes deployment in open policy regimes, and alters partner choice before payoff clearly moves. Under abrupt betrayal, however, the same mechanism can sharpen action around outdated assumptions and reduce payoff. Extending affective active inference to social settings therefore requires localizing precision over partner models and treating affect as a deployment signal that can both support and misdirect social action.

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